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Total Synthesis | *Very Important Paper* |

Unexpected Selectivity of Intramolecular [3+2] Cycloaddition of Trimethylenemethane (TMM) Diyl toward Total Synthesis of Conidiogenone B

Myungjo J. Kim,[a,b] Sanghyeon Lee,[a] Taek Kang,[a] Mu-Hyun Baik,*[b,a] and Hee-Yoon Lee*[a]

Abstract: In the course of a study toward a total synthesis of conidiogenone B through TMM diyl-mediated tandem cycloaddition reaction, unexpected regioselectivity for a bridged isomer was observed. The synthetic and computational studies with model compounds revealed that regioselectivity can be rendered by changing substitution patterns on the TMM diyl moiety and the olefin moiety by altering the activation energy of cycloaddition reaction through a singlet TMM diyl pathway.

Introduction

Total synthesis of natural products with complex structural features, especially with contiguous all-carbon quaternary centers, have challenged synthetic chemistry community since the dawn of the 20th century[1] and triggered development of various synthetic methodologies and strategies.[2] Among synthetic strategies for the construction of structurally congested natural products, trimethylenemethane (TMM) diyl-mediated tandem cycloaddition reaction proved particularly attractive, as it has offered efficient means to construct all-carbon quaternary centers at ring junctions and rapid access to architectural complexity through the tandem process.[3] From the viewpoint of "ideal synthesis",[4] this cascade process involving the successive intramolecular [3+2] cycloaddition reactions to form three rings at once is attractive because it is highly step-, and pot-economical. We have successfully applied this synthetic methodology to the total synthesis of highly congested natural products possessing triquinane structure (Figure 1).

Total syntheses of various triquinanes were executed through TMM diyl [3+2] cycloaddition reaction, and complex natural products with multiple quaternary centers were successfully synthesized through the TMM diyl-mediated tandem cycloaddition reaction. Following these successful examples, we were to extend the synthetic strategy to the total synthesis of another challenging structure of tetracyclic diterpene, conidiogenone B,[5] though total synthesis of conidiogenone B did not appear straightforward through the tandem cycloaddition reaction via TMM diyl [3+2] cycloaddition reaction.[6]

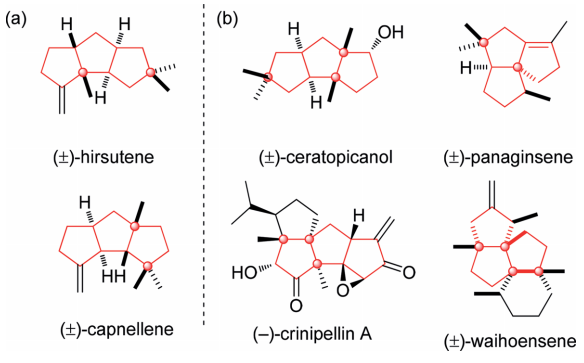


Figure 1. Complex natural products synthesized by (a) TMM diyl [3+2] cycloaddition reaction[7] and (b) TMM diyl-mediated tandem cycloaddition reaction.[8] (Red circles denote quaternary carbon centers.).

It was anticipated that a known synthetic intermediate (**7**)[6] of conidiogenone B (**8**) could be prepared from 6 by introducing the *gem*-dimethyl quaternary center. The tetracyclic compound **6** was expected to be obtained from 5 through the TMM diyl-mediated tandem cycloaddition reaction (Figure 2).

Results and Discussion

The synthesis commenced with chemoselective protection of commercially available Wieland–Miescher ketone **1** following the known procedure[9] (Scheme 1). Nucleophilic epoxidation of the resulting enone **9** afforded **10**, and subsequent Eschenmoser–Tanabe fragmentation resulted in γ-ethynyl ketone **11**. Introduction of an *exo*-cyclic methylene to the ketone moiety by Wittig reaction cleanly produced **3** in 78 %. Gratifyingly, **3** was successfully coupled with **4** generated from **12** in situ in the presence of CuI and LiO*t*Bu in 1,4-dioxane,[10] to yield allene

[a] *Department of Chemistry, Korea Advanced Institute of Science and Technology (KAIST)*
Daejeon 34141, Republic of Korea
E-mail: hylee@kaist.ac.kr
http://chem.kaist.ac.kr
[b] *Center for Catalytic Hydrocarbon Functionalizations; Institute for Basic Science (IBS)*
Daejeon 34141, Republic of Korea
E-mail: mbaik2805@kaist.ac.kr
http://cchf.ibs.re.kr

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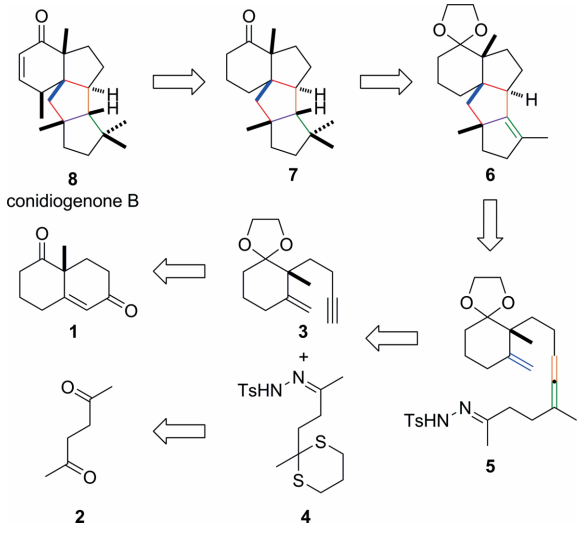
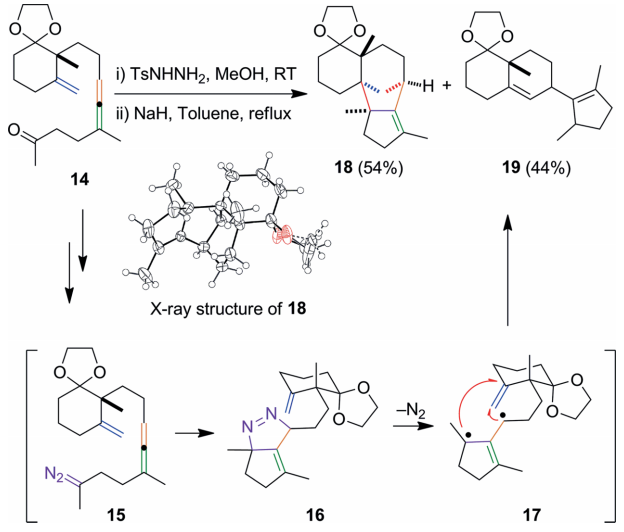
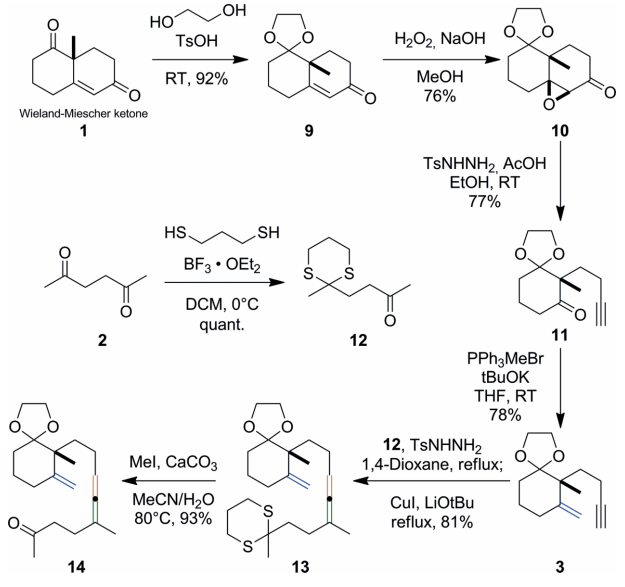


Figure 2. Retrosynthetic analysis of conidiogenone B.



Scheme 2. Tandem cycloaddition of **14**.

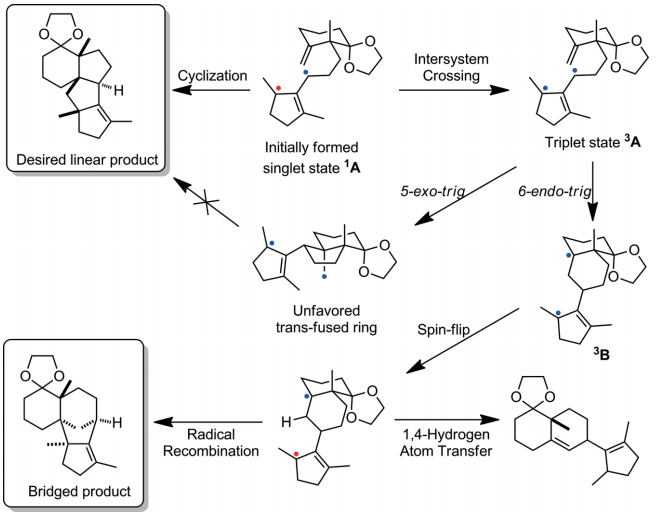
13 in 81 %. Deprotection of the dithiane moiety quantitatively afforded the ketone **14** that possessed all the carbon units for the tetracyclic structure of conidiogenone B.



Scheme 1. Preparation of cycloaddition precursor **14**.

When the sodium salt of hydrazone compound **5** was heated to NaH, formation of **6** was expected through TMM diyl **17** via **15** and **16** (Scheme 2). While the intramolecular [3+2] cycloaddition reaction between the TMM diyl and olefin of **17** proceeded smoothly, to our surprise, the bridged product **18** was obtained as the major product which structure was unambiguously confirmed by single-crystal X-ray analysis.[11] In addition to the major product **18**, 1,4-hydrogen atom transfer (1,4-HAT) product[12] **19** was obtained as a single diastereomer though the relative stereochemistry was not determined.

Both of the products should have formed via radical intermediate ^{3}**B** obtained through the triplet pathway and the formation of the bridged isomer could be understood by the conformational analysis of the five-membered ring formation through the triplet state ^{3}**A**. It would form the unfavorable *trans*-fused ring system if it were to form the triquinane structure and thus the *endo*-cyclic six-membered ring formation would be favored (Scheme 3).

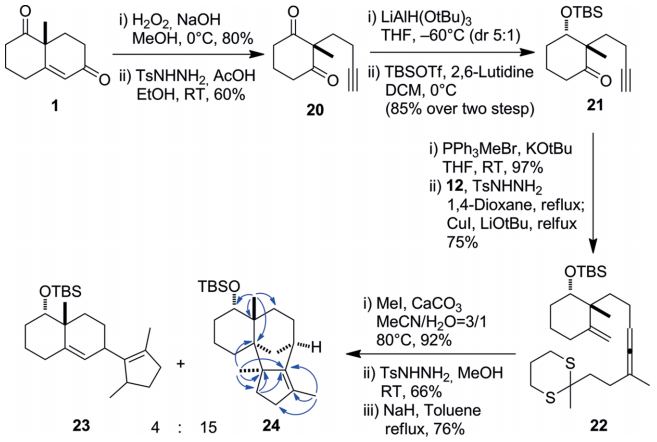


Scheme 3. Mechanistic layout for the generation of linear/bridged cycloaddition products and 1,4-HAT product.

To overcome this presumed unfavorable conformational preference in the intermediate **17**, the ketal functional group of **14** was replaced with an alkoxy group in a hope that the change of conformational preference could favor the formation of the *cis* fused five membered ring. The modified synthesis commenced with a direct nucleophilic epoxidation and Eschenmoser–Tanabe fragmentation of **1**. (Scheme 4) The resulting symmetric diketone **20** is subjected to $LiAlH(OtBu)_3$[13] and TBSOTf to afford protected alcohol **21** in 85 % with *dr* 5:1. Then alkynone **21** was subjected to the same reaction sequence as the sequence for the synthesis of **18**. The tandem cycloaddition reaction, contrary to our hope, again produced the bridged cycloaddition product **24** along with an interrupted cyclization product **23**. These results were quite surprising since the com-

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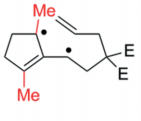
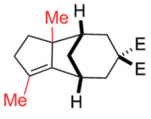
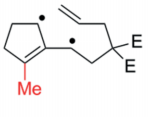
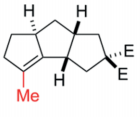
plete selectivity for the bridged cycloaddition product has not been observed before.



Scheme 4. Modified synthetic route of tandem cycloaddition reaction. (Blue arrows represent key HMBC correlations.).

To understand the source of the selectivity, the cycloaddition reaction of model system was studied as shown in Table 1. Effect of another methyl group on the TMM diyl was compared to the one with only one methyl group on the TMM diyl. Intramolecular [3+2] cycloaddition of the dimethyl-substituted TMM diyl **25** resulted in the exclusive formation of the bridged product **26**. In comparison, monomethyl-substituted TMM diyl **27**, showed the exclusive production of linear product **28**. This result clearly indicated that selectivity for the bridged product was greatly influenced by the presence of the second methyl group on the TMM diyl. Formation of the bridged cycloaddition product from TMM diyl cycloaddition reaction has been reported in the past by changing steric environment on the olefin moiety, and different electronic and steric environment for the cycloaddition reaction altered the selectivity (Table 2).[14]

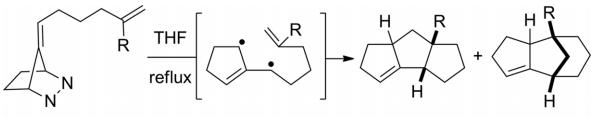
Table 1. Intramolecular [3+2] TMM diyl-olefin cycloadditions of model TMM diyls.[a]

Entry	TMM Diyl	Cycloaddition Result	Yield
1			57%
	25	**26**[b]	
2			88%
	27	**28**	

[a] Methyl groups are highlighted in red. [b] Relative stereochemistry of the bridgehead methyl group is not determined. (E = COOEt).

Electron withdrawing acetyl-substituted **29a** showed a regioselectivity for the linear isomer, while alkyl-substituted **29b** and **29c** afforded fair amount of both isomers. In the case of **29d** and **29e** with sterically demanding alkyl substituents, the bridged isomers became the dominant product. This observation was explained by the fact that singlet TMM diyl only forms linearly fused product while triplet TMM diyl could form linear products as well as bridged products. Thus, stabilizing the $LUMO_{Olefin}$ would render the $HOMO_{TMM}$–$LUMO_{Olefin}$ gap smaller to make the singlet cycloaddition faster than intersystem crossing into the triplet TMM diyl, so that a linear isomer is preferably generated.[15] When we calculated the $HOMO_{TMM}$–$LUMO_{Olefin}$ gap for Little's report, the result showed a clear effect of the electron withdrawing group (EWG) as the energy gap was significantly smaller than others (Table 2, entry 1). However, for the TMM diyls with sterically different environment, $HOMO_{TMM}$–$LUMO_{Olefin}$ gap did not provide any information for the product ratio (Table 2, entries 2–5). The selectivity observed in Table 1 could not be explained either as the $HOMO_{TMM}$–$LUMO_{Olefin}$ gap did not show much difference in electronic environment.

Table 2. Regioselectivity for the reported TMM diyls[14] and our model TMM diyls, alongside their calculated $HOMO_{TMM}$–$LUMO_{Olefin}$ gaps.



Entry	TMM Diyl	R	Linear/ Bridged	Yield	$HOMO_{TMM}$–$LUMO_{Olefin}$ Gap[a]
1	**30a**	Ac	7.3:1	98 %	3.183 eV
2	**30b**	$–CH_2OH$	1:1.9	91 %	4.846 eV
3	**30c**	$–CH(OMe)_2$	1:2.0	86 %[b]	5.031 eV
4	**30d**	$–CMe(OMe)_2$	1:15.7	68 %[c]	4.793 eV
5	**30e**	$–CMe(OCH_2CH_2O)$	1:10	89 %	
6	**25**		bridged only	57 %	
7	**27**		linear only	88 %	

[a] See **Supporting Information** for computational details. [b] Yield over two steps: acetalization and cycloaddition. [c] Yield over two steps: cycloaddition and ketal deprotection.

At this point, as we verified that the frontier molecular orbital (FMO) consideration cannot account for other factors than electronic factors on olefins and TMM diyls, it would be valuable to have more elaborate means that could explain the outcome of the cycloaddition reaction and could be capable of predicting selectivity of the reaction. Therefore, we calculated the actual cycloaddition barriers of singlet TMM diyls,[16] and compare the barriers with the corresponding regioselectivity outcomes. Gibbs-free energy barriers ($\Delta G_{sol}^{\ddagger}$) of singlet cycloadditions are calculated to include entropy, zero-point energy, and solvation corrections as well as the steric influence (Table 3). [M06/cc-pVTZ(-f)//M06/6-31G**].

The barrier of singlet TMM diyl cycloaddition for **30a** was estimated to be significantly lower than that of **30d** by 6.6 kcal/mol (Table 3). This considerable difference in transition state energy should be attributed to both the stabilization of $LUMO_{TMM}$ of **30a** by the acetyl substituent and the steric hindrance from the ketal substituent of **30d**. This difference indicated that compatibility of the cycloaddition through the singlet pathway to the triplet pathway of **30a** could be 6–7 orders of magnitude faster than that of **30d**. Thus, cycloaddition reac-

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Table 3. Comparison between observed regioselectivity and calculated singlet cycloaddition barrier.[a]

Entry	TMM Diyl	Regioselectivity (Linear isomer:Bridged isomer)	Calculated Singlet Cycloaddition Barrier
1	**30d**	1:15.7	11.81 kcal/mol
2	**30a**	7.3:1	5.24 kcal/mol
3	**25**	Bridged isomer only	16.33 kcal/mol
4	**27**	Linear isomer only	11.06 kcal/mol
5	**32**	Linear isomer only (expected)	9.98 kcal/mol

[a] 343.15 K and ε = 7.600 (THF) for calculations of **30a**, **30d**, and their corresponding cycloaddition transition states. 383.15 K and ε = 2.379 (toluene) for calculations of **25**, **27**, **32**, and their corresponding cycloaddition transition states.

tion of **30a** followed mainly through the singlet pathway to produce the linear isomer as the major product[14a] and cycloaddition reaction of **30d** followed the triplet pathway to produce the bridged isomer as the major product due to steric interaction.

In the case of **25** and **27**, cycloaddition barriers were estimated to be 16.3 and 11.1 kcal/mol, respectively, implying that the extra methyl group of **25** played a crucial role to dictate the reaction pathway. This difference of 5.2 kcal/mol is significant and is easy to understand considering the transition state structures shown in Figure 3. The sterically demanding methyl group is very close to the bond forming carbon and leads to a more pronounced structural distortion characterized by a C–C distance of 2.53 Å (**25-TS**), which is notably longer than 2.31 Å that is seen in **27-TS**. Although the cycloaddition is formally synchronous, the optimized transition state structures reveal a significant degree of asynchronous nature with the two proximal C–C distances of 2.78 and 2.93 Å being notably longer than the distal C–C distances. Intrinsic reaction coordinate (IRC) calculations indeed showed a clear shoulder[17] for **27-TS**, which is found for a highly asynchronous process (see **Supporting Information**). Our calculations for the singlet pathway predicted that the carbocyclization of **25** was around 5–6 orders of magnitude slower than that of **27**, and thus facilitated the complete intersystem crossing to triplet and selectively afforded the bridged product **26** due to the steric hindrance imposed by the methyl group.

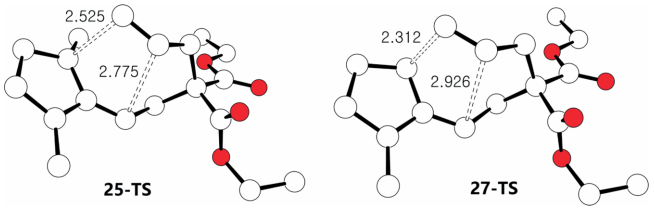
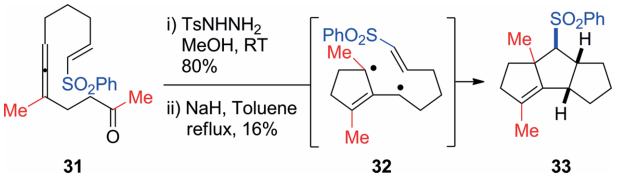


Figure 3. Transition state structures of **25-TS** and **27-TS**. Color notations: Oxygen-red; Carbon-white. Hydrogen atoms of ORTEP are omitted for clarity. [383.15 K and ε = 2.379 (toluene) for calculations.].

With this information, we anticipated that the electronic effect on the olefin moiety might override the steric bias from the extra methyl group on the TMM moiety for the cycloaddition through the singlet pathway dominant. We first conducted a computational study of TMM diyl **32** bearing both an EWG on the olefin moiety and the extra methyl group on TMM diyl part, as well as its singlet cycloaddition transition state **32-TS**.[18] The calculated cycloaddition barrier for **32** was estimated to be 10.0 kcal/mol which is considerably lower than those of **25** and **27** by 6.3 and 1.1 kcal/mol. From this comparison, we anticipated that linear isomer **33** could be obtained from **32** (Scheme 5). To verify the prediction from the calculation, we prepared TMM diyl precursor **31**, and examined the tandem cycloaddition reaction via TMM diyl **32**. The tandem cycloaddition of **31** via TMM diyl **32** indeed resulted in the linear product **33**, albeit in low yield presumably due to the decomposition of the fragile vinyl sulfone. This result assured that addition of an EWG to the olefin moiety can override the selectivity for the bridged isomer arose from the extra methyl group on the TMM diyl moiety.



Scheme 5. Verification of prediction from energy calculation.

These results strongly indicate that preference of intramolecular cycloaddition of a TMM diyl with an olefin could be predicted and controlled, as the regioselectivity is governed by the electronic and steric factors on both the olefin and the TMM diyl. These factors control the kinetics of the intramolecular [3+2] cycloaddition reaction of TMM diyls. When there is low steric influence during the cycloaddition reaction, the linear isomer would be the major product regardless of the spin state of the TMM diyl. When the cycloaddition barrier is lowered by the electronic effect, the singlet TMM diyl would undergo cycloaddition reaction to form exclusively the linear isomer before the intersystem crossing into the triplet TMM diyl would take place. In the case of **32**, it was confirmed that stabilizing the $LUMO_{Olefin}$ could overcome the steric clash from the extra methyl substituent at the TMM diyl. These results hinted that the total synthesis of conidiogenone B through tandem cycloaddition reaction via TMM diyl could be feasible as modification of the substrate for the cycloaddition that should alter the electronic environment of the olefin moiety to shift the reactivity in favor of the desired linear product.

Conclusions

Toward the total synthesis of conidiogenone B, the precursor for the tandem cycloaddition reaction via TMM diyl was prepared from Wieland–Miescher ketone **1** through Eschenmoser–Tanabe fragmentation and Wang allenation. Unexpected regioselectivity of the cycloaddition for a bridged isomer was obtained from the tandem cycloaddition reaction, which led us to explore the intrinsic nature of the electronic and steric effect for selectivity of cycloaddition by synthetic and computational model studies. We obtained some valuable conclusions from the model studies as follows. (1) An extra methyl group on TMM diyl moiety significantly reduces the rate of the singlet cycloaddition reaction and exerted enough steric interaction for the

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exclusive formation of the bridged isomer via triplet pathway. (2) EWG on the olefin moiety renders the cycloaddition through singlet pathway relatively faster than intersystem crossing to produce the linear isomer. (3) This electronic effect by an additional EWG on the olefin moiety could even override the steric interaction from the second extra methyl group on the TMM diyl to yield the linear isomer. It is also notable that there have been only a handful of examples with regioselectivity for a bridged cycloaddition product,[8b,14,19] and the current study reveals that the selectivity between linear isomer and bridged isomer could be controlled by altering the substituents on the TMM diyl and/or the olefin.

Experimental Section

General Information: All reactions were carried out in oven-dried glassware with rubber septa on, under Ar atmosphere. Liquids and solutions were transferred by syringe or cannula, and were introduced through rubber septa. Unless otherwise stated, reactions were conducted at room temperature. Concentration of solutions was accomplished using a Büchi rotary evaporator which was generally followed by removal of residual solvents on a vacuum line held at 0.1–1 Torr. Unless otherwise noted, all commercial reagents and solvents were used without additional purification. Chromatography grade hexane and ethyl acetate were distilled before use; Et_2O and THF were distilled from sodium and benzophenone; dichloromethane was distilled from P_2O_5. Toluene was washed with concentrated sulfuric acid, water and saturated $NaHCO_3$ solution, and then dried with sodium for more than 12 h before distillation. Analytical thin layer chromatography (TLC) was performed on Merck precoated silica gel 60 F254 plates. Visualization on TLC was achieved by use of UV light (254 nm), exposure to iodine vapor, or treatment with acidic anisaldehyde or ceric ammonium molybdate stain followed by heating. Flash column chromatography was carried out using Merck 60, 230–400 mesh ASTM. ^{1}H nuclear magnetic resonance (NMR) spectroscopy was recorded on Bruker Fourier Transform AM 300 (300 MHz), Bruker Avance 400 (400 MHz), Bruker Ascend 400 (400 MHz) or Agilent Technologies DD2 (600 MHz) spectrometers. Chemical shifts were reported in δ units, parts per million (ppm) relative to the singlet as 7.24 ppm for [D]chloroform or 7.16 ppm for [D_6]benzene. The following abbreviations were used to describe peak patterns when appropriate: b = broad, s = singlet, d = doublet, t = triplet, q = quadruplet, m = multiplet. Coupling constant, *J*, was reported in Hertz unit (Hz). ^{13}C NMR was recorded on Bruker Fourier Transform AM 300 (75 MHz), Bruker Avance 400 (100 MHz), Bruker Ascend 400 (100 MHz) or Agilent Technologies DD2 (150 MHz) spectrometers and was fully decoupled by broadband decoupling. Chemical shifts were reported in ppm with the centerline of the triplet for [D]chloroform set at 77.23 ppm or [D_6]benzene set at 128.39 ppm. Mass spectra were obtained on Bruker Daltonik micrOTOF-QII high resolution mass spectrometer using electrospray ionization (ESI) method, and JMS 700 (magenetic sector-electric sector double focusing mass analyzer) using electron ionization (EI) method.

8a-Methyl-3,4,8,8a–tetrahydro-2*H*-spiro[naphthalene-1,2′-[1,3]-dioxolan]-6(7*H*)-one (9):[9] To a solution of ground Wieland–Miescher ketone (1843 mg, 10.134 mmol, 1 equiv.) in ethylene glycol (50.6 mL, 0.2 M) were added 4 Å molecular sieves and *p*-toluenesulfonic acid monohydrate (1928 mg, 10.134 mmol, 1 equiv.) at room temperature. After 23 min at room temperature, the reaction mixture was added to a saturated aqueous sodium bicarbonate solution (200 mL) with ice, and the aqueous layer was extracted with ethyl acetate (3 × 200 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered through silica, and concentrated under reduced pressure. To the resulting crude syrup with a big stir bar in a 1 L round-bottom flask was added a minimal amount of ether. The resulting homogeneous solution was stirred at 0 °C and hexane was added dropwise until the mixture got murky. The flask was kept in the fridge overnight and **9** (2071 mg, 92 %) was obtained as white crystal by filtration with hexane. The mother liquor could be concentrated under reduced pressure and be recycled for more crops on a bigger scale; ^{1}H and ^{13}C NMR spectra are identical to those of the reported literature;[20] HRMS (ESI): *m/z* calcd. for $C_{13}H_{18}O_3$ [M + Na]$^+$ 245.1148, found 245.1148.

(*rel-1aS,4aR,8aS)*-4a-Methyltetrahydro-3*H*,6*H*-spiro[naphtho-[1,8a–b]oxirene-5,2′-[1,3]dioxolan]-2(1aH)-one (10):[21,22] To a solution of **9** (459 mg, 2.067 mmol, 1 equiv.) in methanol (2.67 mL, 0.775 M) were added hydrogen peroxide aqueous solution [30 % (w/w), 0.61 mL, 5.994 mmol, 2.9 equiv.] dropwise and sodium hydroxide aqueous solution [10 % (w/w), 0.36 mL, 0.992 mmol, 0.48 equiv.] dropwise at 0 °C over 30 min. The reaction mixture was stirred at 0 °C for 6 h and in the cold room (1–5 °C) for 3 d. Additional hydrogen peroxide aqueous solution (1 equiv.) was added dropwise after each 24 h period. The reaction mixture was added to saturated sodium thiosulfate aqueous solution (200 mL) (or the remaining hydrogen peroxide could be removed under reduced pressure), and the aqueous layer was extracted with ethyl acetate (3 × 200 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:3) to afford **10** [376 mg, 76 % (BRSM 80 %)]; ^{1}H NMR (400 MHz, [D]Chloroform) δ = 4.04–3.78 (m, 4*H*), 3.07 (s, 1H), 2.31–2.15 (m, 2H), 1.85–1.54 (m, 8H), 1.24 (s, 3H); ^{13}C NMR (101 MHz, [D]Chloroform) δ = 207.69, 113.62, 69.17, 65.50, 64.28, 63.27, 42.86, 34.56, 30.22, 29.60, 25.39, 21.69, 20.20; HRMS (ESI): *m/z* calcd. for $C_{13}H_{18}O_4$ [M + Na]$^+$ 261.1097, found 261.1092.

6-(But-3-yn-1-yl)-6-methyl-1,4-dioxaspiro[4.5]decan-7-one (11): To a stirred solution of **10** (140 mg, 0.589 mmol, 1 equiv.) in ethanol (5.9 mL, 0.1 M) was added *p*-toluenesulfonyl hydrazide at room temperature. After for 3 h, acetic acid (33.7 µL, 0.589 mmol, 1 equiv.) was added dropwise to the resulting mixture at room temperature. After 12 hours, the resulting mixture was added to saturated potassium carbonate aqueous solution (100 mL) and the aqueous layer was extracted with ethyl acetate (3 × 100 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:7) to afford **11** (101 mg, 77 %). Alternative procedure[22] can be used and has been demonstrated in our laboratory as follows. To a solution of **10** (108 mg, 0.453 mmol, 1 equiv.) in dichloromethane (0.36 mL, 1.26 M) were added *p*-toluenesulfonyl hydrazide (89 mg, 0.462 mmol, 1 equiv.) and acetic acid (34 µL, 0.589 mmol, 1.35 equiv.) dropwise at 0 °C. After being stirred for 30 min at 0 °C and then 1.5 h at room temperature, sodium carbonate (115 mg, 1.359 mmol, 3 equiv.) and methanol (0.54 mL, final concentration 0.5 M) were added to the reaction mixture. The resulting solution was stirred at 40 °C for 2 h and was poured into saturated sodium bicarbonate aqueous solution (100 mL). The aqueous layer was extracted with dichloromethane (3 × 100 mL) and the combined organic layers were washed with water (100 mL). The washed combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chroma-

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tography (silica gel, ethyl acetate/hexane = 1:7) to afford **11** (71 mg, 71 %); ^{1}H NMR (400 MHz, [D$_6$]benzene) δ = 3.29–3.26 (m, 4H), 2.21–2.12 (m, 1H), 2.12–1.89 (m, 5H), 1.78 (t, *J* = 2.3 Hz, 1H), 1.62–1.51 (m, 1H), 1.51–1.40 (m, 1H), 1.40–1.31 (m, 1H), 1.31–1.21 (m, 1H), 1.04 (s, 3H); ^{13}C NMR (101 MHz, [D$_6$]benzene) δ = 209.76, 114.01, 84.75, 69.39, 65.48, 65.34, 58.87, 37.36, 34.53, 29.98, 19.83, 14.57, 14.26. HRMS (ESI): *m/z* calcd. for $C_{13}H_{18}O_3$ [M + Na]$^+$ 245.1148, found 245.1150.

6-(But-3-yn-1-yl)-6-methyl-7-methylene-1,4-dioxaspiro[4.5]decane (3): Prior to the reaction set-up, methyltriphenylphosphonium bromide (1410 mg, 3.867 mmol, 3 equiv.) was stirred and heated at 110 °C overnight. To a solution of dry methyltriphenylphosphonium bromide in THF (5 mL) was added potassium *tert*-butoxide (434 mg, 3.867 mmol, 3 equiv.) and the resulting solution was stirred at 0 °C for 30 min, followed by the addition of **11** (287 mg, 1.289 mmol, 1 equiv.) in THF (1.4 mL, final concentration 0.2 M) via cannula at 0 °C. After being stirred for 24 h at room temperature, the resulting mixture was added to saturated ammonium chloride aqueous solution (150 mL) and the aqueous layer was extracted with ethyl acetate (3 × 150 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:20) to afford **3** (223 mg, 78 %); ^{1}H NMR (400 MHz, [D]Chloroform) δ = 4.87 (s, 1H), 4.67 (s, 1H), 3.94–3.85 (m, 4H), 2.20–2.06 (m, 2H), 2.04–1.93 (m, 1H), 2.04–1.93 (m, 1H), 1.92–1.71 (m, 4H), 1.70–1.55 (m, 2H), 1.53–1.38 (m, 1H), 0.97 (s, 3H); ^{13}C NMR (101 MHz, [D]Chloroform) δ = 150.71, 113.00, 110.86, 85.30, 68.05, 65.38, 65.19, 49.23, 33.76, 31.86, 30.27, 23.54, 15.57, 13.47; HRMS (ESI): *m/z* calcd. for $C_{14}H_{20}O_2$ [M + Na]$^+$ 243.1356, found 243.1345.

4-(2-Methyl-1,3-dithian-2-yl)butan-2-one (12): To a solution of hexa-2,5-dione (0.286 mL, 1.972 mmol, 2 equiv.) and 1,3-propanedithiol (0.1 mL, 0.986 mmol, 1 equiv.) in dichloromethane (9.9 mL, 0.1 M) was added boron trifluoride diethyl etherate (0.262 mL, 0.986 mmol, 1 equiv.) dropwise at 0 °C. The reaction mixture was stirred for 30 min at 0 °C and was added to sodium bicarbonate aqueous solution (200 mL). The aqueous layer was extracted with dichloromethane (3 × 200 mL), and the combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:7) to afford **12** (214 mg, quant.); ^{1}H and ^{13}C NMR spectra are identical to those of the reported literature.[23] HRMS (ESI): *m/z* calcd. for $C_9H_{16}OS_2$ [M + Na]$^+$ 227.0535, found 227.0535.

6-Methyl-6-[5-methyl-7-(2-methyl-1,3-dithian-2-yl)hepta-3,4-dien-1-yl]-7-methylene-1,4-dioxaspiro[4.5]decane (13): To a solution of **12** (205 mg, 1.004 mmol, 2.7 equiv.) in 1,4-dioxane (1 mL) were added *p*-toluenesulfonyl hydrazide (200 mg, 1.042 mmol, 2.8 equiv.) and 4 Å molecular sieves at room temperature. The reaction mixture was refluxed for 12 h and was cooled to room temperature. To the resulting solution were added **3 via** cannula (81.9 mg, 0.372 mmol, 1 equiv.) in 1,4-dioxane (5 mL, final concentration 0.08 M), copper(I) iodide (73 mg, 0.383 mmol, 1 equiv.), additional 4 Å molecular sieves, and lithium *tert*-butoxide (164 mg, 2.046 mmol, 5.5 equiv.). The resulting solution was refluxed for 24 h and was cooled down to room temperature to be filtered through a silica pad. The filtrate was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:30 → 1:20) to afford **13** (123 mg, 81 %). (Two-pot procedure of preparing the hydrazone individually is also viable, however in this case one-pot procedure gave better yields most of the time in our laboratory.[10]); ^{1}H NMR (400 MHz, [D]Chloroform) δ = 5.05–5.02 (m, 1H), 4.87 (s, 1H), 4.69 (d, *J* = 1.7 Hz, 1H), 3.97–3.84 (m, 4H), 2.87–2.82 (m, 2H), 2.82–2.72 (m, 2H), 2.18–2.09 (m, 2H), 2.09–1.98 (m, 4H), 1.98–1.88 (m, 2H), 1.88–1.82 (m, 1H), 1.82–1.70 (m, 2H), 1.70–1.65 (m, 1H), 1.70–1.65 (m, 3H), 1.65–1.60 (m, 1H), 1.60–1.52 (m, 2H), 1.57 (s, 3H), 1.53–1.43 (m, 1H), 1.01 (s, 3H); ^{13}C NMR (101 MHz, [D]Chloroform) δ = 200.72, 200.70, 151.60, 151.57, 113.44, 110.31, 91.95, 65.43, 65.41, 65.23, 49.47, 49.15, 39.51, 39.47, 34.47, 34.17, 32.04, 30.33, 30.32, 29.19, 29.14, 28.01, 27.99, 26.71, 25.51, 25.50, 24.29, 24.04, 23.74, 20.03, 19.99, 15.94, 15.92; HRMS (ESI): *m/z* calcd. for $C_{23}H_{36}O_2S_2$ [M + Na]$^+$ 431.2049, found 431.2052.

5-Methyl-9-(6-methyl-7-methylene-1,4-dioxaspiro[4.5]decan-6-yl)nona-5,6-dien-2-one (14): To a solution of **13** (1368 mg, 3.348 mmol, 1 equiv.) in MeCN/H$_2$O = 3:1 (v/v) mixture (36 mL, 0.1 M) were added calcium carbonate (1460 mg, 14.416 mmol, 4 equiv.) and iodomethane (2.25 mL, 36.036 mmol, 10 equiv.) at room temperature. The reaction mixture was heated to 80 °C and was stirred for 4 h. The resulting mixture was cooled down to be added to saturated ammonium chloride aqueous solution (250 mL) and the aqueous layer was extracted with ethyl acetate (3 × 250 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:5) to afford **14** (989 mg, 93 %); ^{1}H NMR (400 MHz, [D]Chloroform) δ = 5.02–4.98 (m, 1H), 4.85 (s, 1H), 4.66 (s, 1H), 3.99–3.79 (m, 4H), 2.49 (t, *J* = 7.2 Hz, 2H), 2.18–2.04 (m, 4H), 2.10 (s, 3H), 1.90–1.71 (m, 3H), 1.67–1.56 (m, 3H), 1.64 (d, *J* = 2.9 Hz, 3H), 1.55–1.41 (m, 2H), 0.98 (s, 3H); ^{13}C NMR (101 MHz, [D]Chloroform) δ = 208.59, 200.51, 200.48, 151.45, 113.36, 110.30, 110.28, 99.14, 99.11, 92.54, 65.39, 65.19, 49.41, 49.39, 41.82, 41.80, 34.20, 34.17, 31.97, 30.29, 30.27, 30.24, 30.21, 27.69, 24.18, 24.13, 23.67, 19.88, 19.85, 15.88; HRMS (ESI): *m/z* calcd. for $C_{20}H_{30}O_3$ [M + Na]$^+$ 341.2087, found 341.2087.

(*rel*-4′*R*,6a′*R*,10a′*S*,10b′*S*)-3′,6a′,10b′-Trimethyl-2′,4′,5′,6′,6a′,9′,10′,10b′-octahydro-1′*H*,8′*H*-spiro[[1,3]dioxolane-2,7′-[[4,10a]-methanobenzo[*e*]azulene] (18) and 6-(2,5-Dimethylcyclopent-1-en-1-yl)-8a-methyl-3,4,6,7,8,8a-hexahydro-2*H*-spiro[naphthalene-1,2′-[1,3]dioxolane] (19): To a solution of **14** (120 mg, 0.376 mmol, 1 equiv.) in methanol (0.75 mL, 0.5 M) was added *p*-toluenesulfonyl hydrazide (87 mg, 0.451 mmol, 1.2 equiv.) at room temperature. After being stirred for 1.5 h at room temperature, the reaction mixture was concentrated under reduced pressure and was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:4 → 1:3) to afford hydrazone precursor (183 mg, quantitative). To a suspension of sodium hydride [60 % (w/w) dispersion in mineral oil, 18.8 mg, 0.470 mmol, 1.25 equiv.] in toluene (44 mL) was added the freshly made hydrazone precursor (183 mg, 0.376 mmol, 1 equiv.) in toluene (10 mL, final concentration 0.007 M) by cannula at 0 °C. The resulting solution was degassed by Ar balloon at 0 °C for 15 min. The reaction mixture was refluxed for 24 h and was cooled down to be added to saturated ammonium chloride aqueous solution (100 mL). The aqueous layer was extracted with ethyl acetate (3 × 100 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, diethyl ether/hexane = 1:150) to afford **18** (60 mg, 54 %) and 1,4-hydrogen atom transfer product **19** (49 mg, 44 %). The structure of **18** has been unambiguously confirmed by X-ray crystal analysis as shown in Scheme 2 and Figure S1. Refer to the GC–MS data for the purity of **19** (see Supporting Information); [**18**]: ^{1}H NMR (400 MHz, [D$_6$]benzene) δ = 3.60–3.41 (m, 4H), 3.13 (d, *J* = 11.8 Hz, 1H), 2.97–2.84 (m, 1H), 2.76 (s, 1H), 2.61 (dddd, *J* = 11.7, 10.2, 8.5, 1.5 Hz, 1H), 2.17 (dd, *J* = 15.0, 8.3 Hz, 1H), 2.04–1.93 (m, 2H), 1.82 (ddd, *J* = 11.8, 5.4, 2.0 Hz, 1H),

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1.64 (m, 3H), 1.59–1.41 (m, 6H), 1.38 (s, 3H), 1.10 (d, J = 1.3 Hz, 3H); ^{13}C NMR (101 MHz, [D$_6$]benzene) δ = 156.44, 123.03, 116.19, 65.36, 63.53, 62.73, 51.46, 47.18, 40.90, 39.97, 38.30, 33.22, 32.18, 31.86, 29.81, 28.86, 25.55, 22.53, 20.85, 15.13; HRMS (EI): *m/z* calcd. for $C_{20}H_{30}O_2$ [M]$^+$ 302.2240, found 302.2243; [**19**]: ^{1}H NMR (400 MHz, [D$_6$]benzene) δ = 5.88 (d, J = 1.7 Hz, 1H), 3.63–3.44 (m, 4H), 2.95–2.84 (m, 1H), 2.67 (t, J = 1.9 Hz, 2H), 2.30–2.19 (m, 2H), 2.11 (s, 2H), 2.07 (m, 3H), 1.98 (s, 3H), 1.68–1.49 (m, 6H), 1.39 (s, 4H), 1.30–1.27 (m, 3H); ^{13}C NMR (101 MHz, [D$_6$]benzene) δ = 151.14, 144.44, 144.26, 137.19, 136.88, 125.60, 113.44, 65.30, 65.22, 44.91, 42.42, 40.57, 32.79, 31.52, 30.66, 30.43, 28.68, 27.59, 27.21, 23.64, 21.29, 18.12, 16.61, 14.96; MS (Low EI): *m/z* calcd. for $C_{20}H_{30}O_2$ [M]$^+$ 302, found 302.

2-(But-3-yn-1-yl)-2-methylcyclohexane-1,3-dione (20): To a solution of Wieland–Miescher ketone (2.0 g, 10.997 mmol, 1 equiv.) in methanol (55 mL, 0.2 M) were added hydrogen peroxide aqueous solution [35 % (w/w), 2.8 mL, 32.991 mmol, 3 equiv.] and sodium hydroxide aqueous solution [10 % (w/w), 2.0 mL, 5.498 mmol, 0.5 equiv.] dropwise at 0 °C. The resulting mixture was stirred for 12 h at 0 °C and was added to saturated sodium thiosulfate aqueous solution (200 mL) (or the remaining hydrogen peroxide could be removed under reduced pressure). The aqueous layer was extracted with ethyl acetate (3 × 200 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:3 → 1:2) to afford nucleophilic epoxidation product (1700 mg, 80 %). This procedure has been repeated in 2.9 gram scale and yielded 2262 mg (72 %) which was used in the following procedure, alongside 1700 mg of the product from the previous batch. To a solution of the prepared epoxide precursor [3962 mg (1700 mg + 2262 mg), 20.398 mmol, 1 equiv.] in ethanol (81.6 mL, 0.25 M) was added *p*-toluenesulfonyl hydrazide (4001 mg, 20.806 mmol, 1.02 equiv.), and the resulting solution was stirred for 3 h at room temperature followed by the addition of acetic acid (1.18 mL, 20.398 mmol, 1 equiv.) at room temperature. The reaction mixture was stirred for 12 h at room temperature and was added to saturated potassium carbonate aqueous solution, and the aqueous layer was extracted with ethyl acetate (3 × 200 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:4) to afford **20** (2189 mg, 60 %); ^{1}H NMR (400 MHz, [D$_6$]benzene) δ = 2.05 (ddd, J = 7.2, 5.8, 4.4 Hz, 4H), 2.00–1.94 (m, 2H), 1.94–1.88 (m, 2H), 1.80 (t, J = 2.5 Hz, 1H), 1.25–1.06 (m, 2H), 0.99 (s, 3H); ^{13}C NMR (101 MHz, [D$_6$]benzene) δ = 208.85, 83.86, 70.40, 64.86, 37.99, 35.07, 21.23, 17.95, 14.86; HRMS (ESI): *m/z* calcd. for $C_{11}H_{14}O_2$ [M + Na]$^+$ 201.0886, found 201.0885.

(*rel*-2*S*,3*S*)-2-(But-3-yn-1-yl)-3-[(*tert*-butyldimethylsilyl)oxy]-2-methylcyclohexan-1-one (21): To a solution of **20** (510 mg, 2.863 mmol, 1 equiv.) in THF (19 mL, 0.15 M) was added lithium tri-*tert*-butoxyaluminum hydride (LTBA, 1 M solution in THF, 3.58 mL, 3.578 mmol, 1.25 equiv.) at –60 °C. After 1 h at –60 °C, 1 N hydrochloric acid (19 mL) was added to the reaction mixture and the resulting solution was extracted with ethyl acetate (3 × 150 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was used in the following procedure without further purification. To a solution of the crude ketol precursor (1 equiv.) in dichloromethane (14.3 mL, 0.2 M) were added 2,6-lutidine (0.67 mL, 5.725 mmol, 2 equiv.) and *tert*-butyldimethylsilyl trifluoromethanesulfonate (1.1 mL, 4.580 mmol, 1.6 equiv.) at 0 °C. After being stirred for 30 min, the reaction mixture was added to saturated ammonium chloride aqueous solution (150 mL), and the aqueous layer was extracted with ethyl acetate (3 × 150 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:30) to afford **21** (719 mg, 85 % over two steps). The relative stereochemistry of the siloxy group is determined by NOESY and referring to the previously reported literature.[24] Major stereoisomers of these LTBA ketone reductions have a resulting hydroxy group that is *anti* to the adjacent methyl group, and *syn* to the other adjacent substituent which is bigger than the methyl group. This has also been verified by a parallel $NaBH_4$ reduction experiment referring to the previously reported literature,[25] which has shown the same stereochemical outcome with the LTBA case, but with a lower *dr*; ^{1}H NMR (400 MHz, [D$_6$]benzene) δ = 3.52–3.27 (m, 1H), 2.18–1.89 (m, 6H), 1.77 (s, 1H), 1.72–1.49 (m, 1H), 1.44 (t, J = 6.0 Hz, 2H), 1.21–1.00 (m, 1H), 0.97 (s, 3H), 0.90 (s, 9H), –0.07 (d, J = 8.9 Hz, 6H); ^{13}C NMR (101 MHz, [D$_6$]benzene) δ = 211.18, 84.86, 84.65, 78.67, 76.91, 69.37, 55.18, 54.85, 37.95, 37.81, 35.46, 32.04, 29.77, 26.32, 26.30, 20.86, 20.83, 19.37, 18.55, 18.29, 14.36, 14.01, –3.83, –3.87, –4.61, –4.63; HRMS (ESI): *m/z* calcd. for $C_{17}H_{30}O_2Si$ [M + Na]$^+$ 317.1907, found 317.1908.

***tert*-Butyldimethyl({(*rel*-1*S*,2*R*)-2-methyl-2-[5-methyl-7-(2-methyl-1,3-dithian-2-yl)hepta-3,4-dien-1-yl]-3-methylenecyclohexyl}oxy)silane (22):** Prior to the reaction set-up, methyltriphenylphosphonium bromide (4347 mg, 11.927 mmol, 3 equiv.) was stirred and heated at 110 °C overnight. To a solution of dry methyltriphenylphosphonium bromide in THF (6 mL) was added potassium *tert*-butoxide (1340 mg, 11.927 mmol, 3 equiv.) and the resulting solution was stirred at 0 °C for 30 min, followed by the addition of **21** (1171 mg, 3.976 mmol, 1 equiv.) in THF (4 mL, final concentration 0.4 M) via cannula at 0 °C. After stirring for 24 h at room temperature, the resulting mixture was added to saturated ammonium chloride aqueous solution (250 mL) and the aqueous layer was extracted with ethyl acetate (3 × 250 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, hexane) to afford methylenated product (1125 mg, 97 %). To a solution of 12 (2043 mg, 9.998 mmol, 2.6 equiv.) in 1,4-dioxane (10 mL, 1 M) were added *p*-toluenesulfonyl hydrazide (1993 mg, 10.383 mmol, 2.7 equiv.) and 4 Å molecular sieves at room temperature. The reaction mixture was refluxed for 12 h and was cooled to room temperature. To the resulting solution were added the prepared methylenated product (1125 mg, 3.845 mmol, 1 equiv.) in 1,4-dioxane via cannula (38 mL, final concentration 0.08 M), copper(I) iodide (754 mg, 3.960 mmol, 1 equiv.), additional 4 Å molecular sieves, and lithium *tert*-butoxide (1700 mg, 21.148 mmol, 5.5 equiv.). The resulting solution was refluxed for 24 h and was cooled down to room temperature to be filtered through a silica pad. The filtrate was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:50 → 1:30) to afford 22 (1380 mg, 75 %); ^{1}H NMR (400 MHz, [D$_6$]benzene) δ = 5.31–5.16 (m, 1H), 4.83 (d, J = 1.9 Hz, 1H), 4.74 (d, J = 1.9 Hz, 1H), 3.51–3.26 (m, 1H), 2.51 (m, 2H), 2.44–2.35 (m, 2H), 2.25 (d, J = 4.3 Hz, 4H), 2.14–1.92 (m, 4H), 1.92–1.74 (m, 4H), 1.70 (dd, J = 11.7, 2.8 Hz, 3H), 1.62 (m, 2H), 1.57 (s, 3H), 1.55–1.50 (m, 2H), 1.15 (s, 3H), 1.04–0.95 (m, 9H), 0.09 to –0.02 (m, 6H); ^{13}C NMR (101 MHz, [D$_6$]benzene) δ = 201.64, 201.60, 153.36, 153.33, 152.02, 138.76, 115.27, 110.23, 110.19, 110.06, 100.15, 99.92, 93.03, 92.99, 79.80, 49.60, 46.70, 46.63, 40.49, 40.47, 33.07, 31.54, 31.49, 31.34, 30.75, 29.92, 29.80, 28.51, 26.97, 26.94, 26.92, 26.81, 26.57, 26.52, 25.90, 25.50, 24.69, 24.25, 22.44, 22.42, 20.29, 20.26,

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18.74, –3.48, –3.51, –4.36; HRMS (ESI): *m/z* calcd. for $C_{27}H_{48}OSiS_2$ [M + Na]$^+$ 503.2808, found 503.2813.

***tert*-Butyl{[(*rel*-1*S*,8a*R*)-6-(2,5-dimethylcyclopent-1-en-1-yl)-8a-methyl-1,2,3,4,6,7,8,8a-octahydronaphthalen-1-yl]oxy}-dimethylsilane (23) and *tert*-Butyldimethyl{[(*rel*-4*R*,6a*R*,7*S*,10a*S*,10b*S*)-3,6a,10b-trimethyl-2,4,5,6,6a,7,8,9,10,10b-decahydro-1*H*-4,10a-methanobenzo[*e*]azulen-7-yl]oxy}silane (24):** To a solution of **22** (1380 mg, 2.870 mmol, 1 equiv.) in MeCN/H_2O = 3:1 (v/v) mixture (28.7 mL, 0.1 M) were added calcium carbonate (1160 mg, 11.479 mmol, 4 equiv.) and iodomethane (1.8 mL, 28.697 mmol, 10 equiv.) at room temperature. The reaction mixture was heated to 80 °C and was stirred for 4 h. The resulting mixture was cooled down to be added to saturated ammonium chloride aqueous solution (200 mL) and the aqueous layer was extracted with ethyl acetate (3 × 200 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel, hexane → ethyl acetate/hexane = 1:20) to afford allenyl ketone product (1031 mg, 92 %). To a solution of the prepared allenyl ketone (437 mg, 1.117 mmol, 1 equiv.) in methanol (3.72 mL, 0.3 M) was added *p*-toluenesulfonyl hydrazide (257 mg, 1.341 mmol, 1.2 equiv.) at room temperature. After being stirred for 1.5 h at room temperature, the reaction mixture was concentrated under reduced pressure and was purified by flash column chromatography (silica gel, ethyl acetate/hexane = 1:10 → 1:5) to afford hydrazone precursor (409 mg, 66 %). To a suspension of sodium hydride [60 % (w/w) dispersion in mineral oil, 36.64 mg, 0.916 mmol, 1.25 equiv.] in toluene (110 mL) was added the freshly made hydrazone precursor (409.2 mg, 0.733 mmol, 1 equiv.) in toluene (12 mL, final concentration 0.006 M) by cannula at 0 °C. The resulting solution was degassed by Ar balloon at 0 °C for 25 min. The reaction mixture was refluxed for 24 h and was cooled down to be added to saturated ammonium chloride aqueous solution (200 mL). The aqueous layer was extracted with ethyl acetate (3 × 200 mL). The combined organic layers were dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting crude mixture was purified by flash column chromatography (silica gel) to afford 1,4-hydrogen atom transfer product **23** (43 mg, 16 %) and 24 (165 mg, 60 %); [**23**]: ^{1}H NMR (400 MHz, [D_6]benzene) δ = 5.94–5.84 (m, 1H), 3.36–3.25 (m, 1H), 2.93–2.80 (m, 1H), 2.68 (s, 2H), 2.29–2.19 (m, 1H), 2.09 (d, J = 1.7 Hz, 3H), 1.99 (s, 3H), 1.69–1.55 (m, 7H), 1.38 (s, 2H), 1.27–1.23 (m, 1H), 1.25 (s, 3H), 1.22–1.12 (m, 2H), 1.02 (s, 9H), 0.10 (d, J = 6.0 Hz, 6H); ^{13}C NMR (101 MHz, [D_6]benzene) δ = 144.43, 144.36, 136.82, 125.67, 80.72, 44.92, 42.22, 39.22, 33.13, 32.38, 31.59, 30.82, 27.88, 27.15, 26.57, 25.18, 25.08, 24.98, 18.77, 16.62, 14.95, –3.30, –4.30; HRMS (ESI): *m/z* calcd. for $C_{24}H_{42}OSi$ [M + Na]$^+$ 397.2897, found 397.2886; [**24**]: ^{1}H NMR (400 MHz, [D_6]benzene) δ = 3.37–3.33 (m, 1H), 3.23 (d, J = 11.6 Hz, 1H), 2.96–2.83 (m, 1H), 2.71 (s, 1H), 2.49 (q, J = 10.9, 10.0 Hz, 1H), 2.19 (dd, J = 14.9, 8.4 Hz, 1H), 2.14–2.04 (m, 1H), 1.95–1.67 (m, 6H), 1.64 (s, 3H), 1.57–1.43 (m, 5H), 1.06 (s, 3H), 1.02 (d, J = 2.4 Hz, 9H), 0.98 (s, 3H), 0.11 (s, 3H), 0.04 (s, 3H); ^{13}C NMR (101 MHz, [D_6]benzene) δ = 156.45, 123.04, 83.78, 62.92, 49.69, 42.46, 40.89, 39.80, 37.65, 36.60, 33.12, 32.26, 31.08, 30.60, 29.51, 27.90, 26.70, 21.90, 19.25, 15.17, –3.96, –4.50; HRMS (EI): *m/z* calcd. for $C_{24}H_{42}OSi$ [M]$^+$ 374.2999, found 374.3006.

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Keywords: Natural products · Total synthesis · Regioselectivity · Radicals · Density functional calculations

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